

NUTRIENTS AND METABOLISM

THE BODY'S INTERNAL BIOCHEMICAL REACTIONS, CHANGES, AND PROCESSES ARE TERMED METABOLISM. DIGESTION PROVIDES THE NUTRIENTS AS RAW MATERIALS, WHICH ENTER METABOLIC PATHWAYS IN ALL CELLS AND TISSUES.

TAKING IN NUTRIENTS

Carbohydrates, fats, proteins, vitamins, minerals, and other nutrients are absorbed at different stages along the digestive tract. Blood from the major absorption sites of the intestines flows along the hepatic portal vein (see p.222) to the liver. Here, nutrients are broken down, stored, and released according to the body's needs.

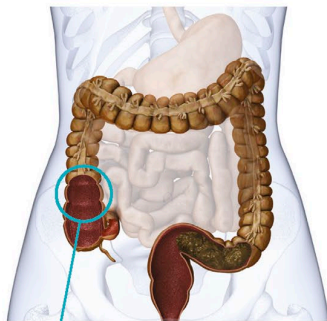


FAT TISSUE

Adipose tissue consists of cells replete with fat droplets that can be used as a concentrated energy store in times of need.

FINAL STAGES OF DIGESTION

The colon absorbs minerals, salts, and some vitamins, and reabsorbs a large amount of water, too. Fibre, such as pectin and cellulose, bulks up the digestive remnants as they are made into faeces. Fibre helps to delay the absorption of some molecules, including sugars, and so spreads their uptake through time rather than in one short "rush". In addition, fibre binds with some fatty substances, such as cholesterol, and helps to prevent their overabsorption.



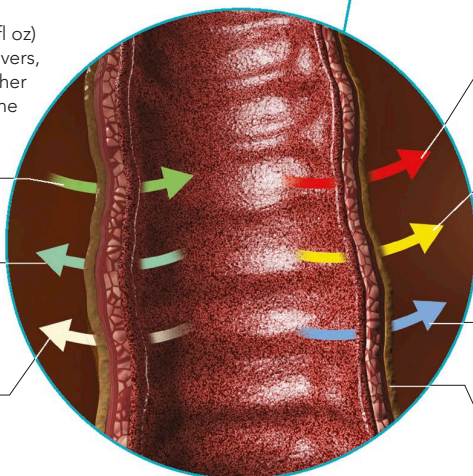
CAECUM

Each day about 100–500ml (3½–17fl oz) of digestive fluids, undigested leftovers, rubbed-off intestinal linings, and other matter enters the first chamber of the large intestine, the caecum.

Bicarbonate and potassium
Secreted into lumen to replace recovered sodium

Chloride
Recovered from faeces; with sodium, it maintains acid-alkali balance in tissues

Sodium
Also recovered from faeces



Vitamin K
Manufactured by symbiotic bacteria

B vitamins
Some types released by bacterial fermentation

Water
Large intestine reabsorbs 2/3 of water in faeces

Colon